

北京邮电大学 2016—2017 学年第 2 学期

《数据库系统原理》期末考试试题(B)

考试 注意 事项	一、学生参加考试须带学生证或学院证明，未带者不准进入考场。学生必须按照监考教师指定座位就坐。 二、书本、参考资料、书包等物品一律放到考场指定位置。 三、学生不得另行携带、使用稿纸，要遵守《北京邮电大学考场规则》，有考场违纪或作弊行为者，按相应规定严肃处理。 四、学生必须将答题内容做在试题答卷上，做在试题及草稿纸上一律无效。 五、填空题用英文答，中文答对得一半分。											
考试 课程	数据库系统 原理	考试时间 2017 年 6 月 20 日										
题号	一	二	三	四	五	六	七	八	九	十	十一	总分
满分	10	16	10	10	6	8	6	11	6	6	11	
得分												
阅卷 教师												

1. Fill in blanks. (1×10 points)

(1) Among the following statements, the correct one/ones is/are BC.

- A. As a type of ~~open-source~~ database systems, SQL Server is developed and distributed by IBM.
- B. MySQL and PostgreSQL are two typical open-source database systems.
- C. The relational model is applicable to managing structured data such as the table data, while XML provides a way to represent semi-structured data, e.g. the data with nested structures.
- D. A on-line shopping site has a three-tier Browser-Server(B/S)-like architecture. Its application programs are programmed in Java, and access Oracle database server via the ~~ODBC~~ interface.
- E. *Big Data* is now a buzzword, and ~~relational~~ databases are able to efficiently manage various types of big data in the forms of tables, texts, web pages, voices, images and videos.

Data Model

(2) The _____ defines the specification of managing data items in database. It is a collection of conceptual tools for describing data structure, data relationships, data semantics, data operations and consistency constraints.

(3) Heap file, sequential file, hashing file organization and clustering file are commonly-used ways of organizing data records in database files. If we first use SQL statement *create table* to create a empty table *instructor*(ID, name, dept-name, salary), assuming that no primary key and index are defined on this table; Then, we add into *instructor* three instructor's information by three sequential *insert into* SQL statements. After these *insert* operations, the data in *instructor* are organized as a heap file on disk.

(4) Query processing refers to the range of activities involved in extracting data from a database. DBMS processes a submitted SQL statement in three sequential steps: parsing and translation, query optimization, and evaluation. In the translation step, the submitted SQL statement is converted to an initial relational algebra expression.

(5) Concurrent schedule S is conflict serializable if it is conflict equivalent to a serial schedule S'.

(6) The designing of database schemas is in accordance with the three-level of data abstract, including conceptual schema design, design, logical and physical schema design.

(7) In two phase locking (2PL) protocol, the ordering of lock point of each transactions in a concurrent schedule S is the same as the ordering of transactions in its equivalent serial schedule S'.

(8) If in a schedule S, a transaction T_j reads a data items previously written by a transaction T_i , the commit operation of T_i appears before (before/after) the commit operation of T_j , the schedule S is recoverable schedule.

(9) Transactions are required to have the ACID properties. durability ensures that, once a transaction has been committed, that transaction's updates do not get lost, even if there is a system failure.

(10) Consider the relation schema $R(A,B,C)$ and the functional dependency set F hold on R . We can compute closure of F as: For each $\gamma \subseteq \{A, B, C\}$, we find the closure γ^+ , and for each $S \subseteq \gamma^+$, we output a functional dependency $\gamma \rightarrow S$.

2. (16 points) Here is the schema diagram for the *Banking* database. The table *branch* describes information on the name, the city located and the assets(资产) of the branches(支行). The customers of the branches are represented as the table *customer*. A customer may have an *account*(存款账户) in a branch. His account is uniquely identified by *account_number*, and *balance* records the amount of money in this account; A customer may also have a *loan*(借款账户) in a branch. His loan is solely identified by *loan_number*, and the amount of money he loans is given by *amount*. The relationships between customers and accounts or loans are modeled as *depositor* or *borrower* respectively.

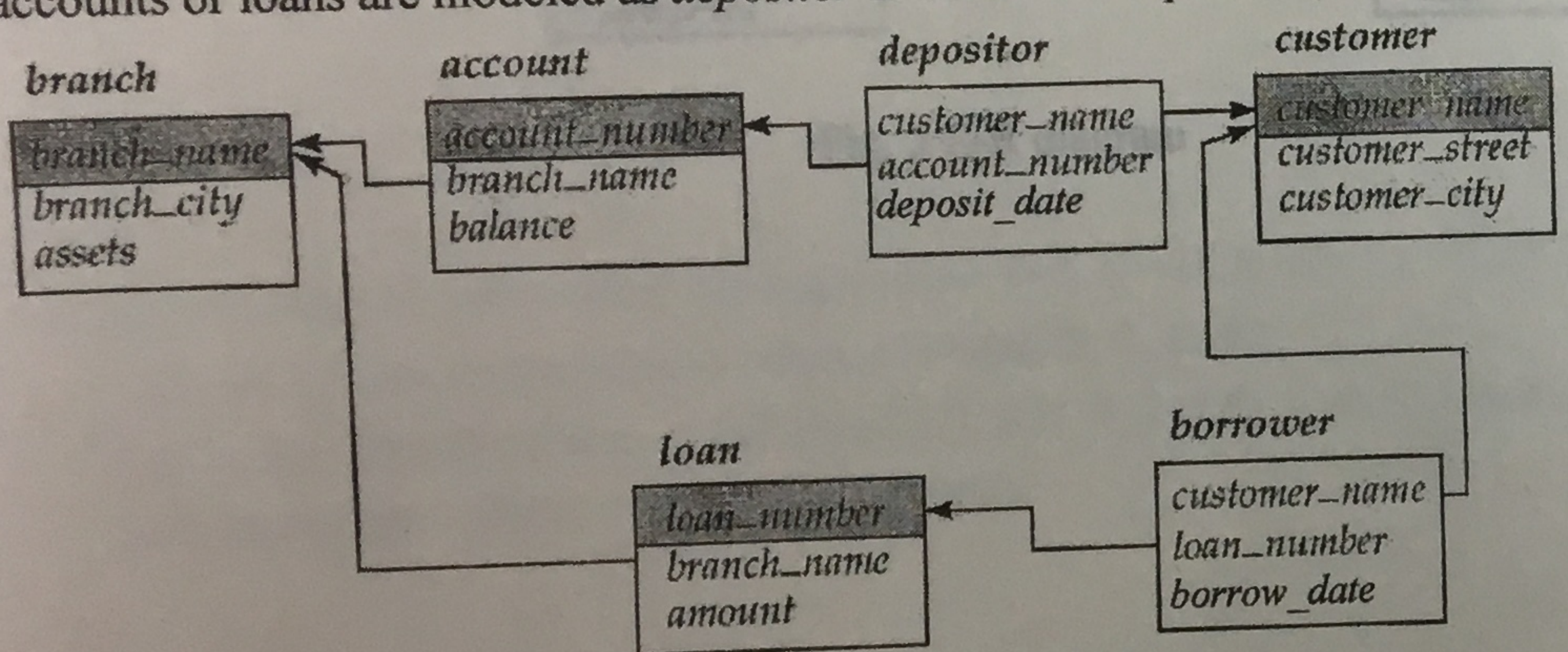


Fig.1 Schema diagram for the *Banking* database

For the following queries, give SQL statements for (1)~(3), and relational algebra expressions for (4)~(5).

(1) (3 points) Create the table *loan*, in which $\{loan_number\}$ is the primary key, and $\{branch_name\}$ is not permitted to be null; there exists a referential integrity constraint from *loan* to *branch*. It is also required that the loan's amount is not below 0.

(2) (3 points) Increase their balance by 5 percentage for all accounts that are opened at the banks locating in *Beijing* and their balances are larger than 10,000.

(3) (5 points) For each branch, find the accounts that are opened in this branch and have the maximum balance among all the accounts in the same branch. For these accounts with maximum balances, list their *account_number*, the names of the branches they locate in and their balances, in a descending order of the balance values.

(4) (2 points) Find the average account balance of every customer. List customer name and average balance.

(5) (3 points) Find the names of all customers who have a loan at the "Beijing Beaty" branch but do not have an account at any branch of the bank.

3. (10 points) Convert the following E-R diagram to the proper relation schemas and identify the primary key of each relation.

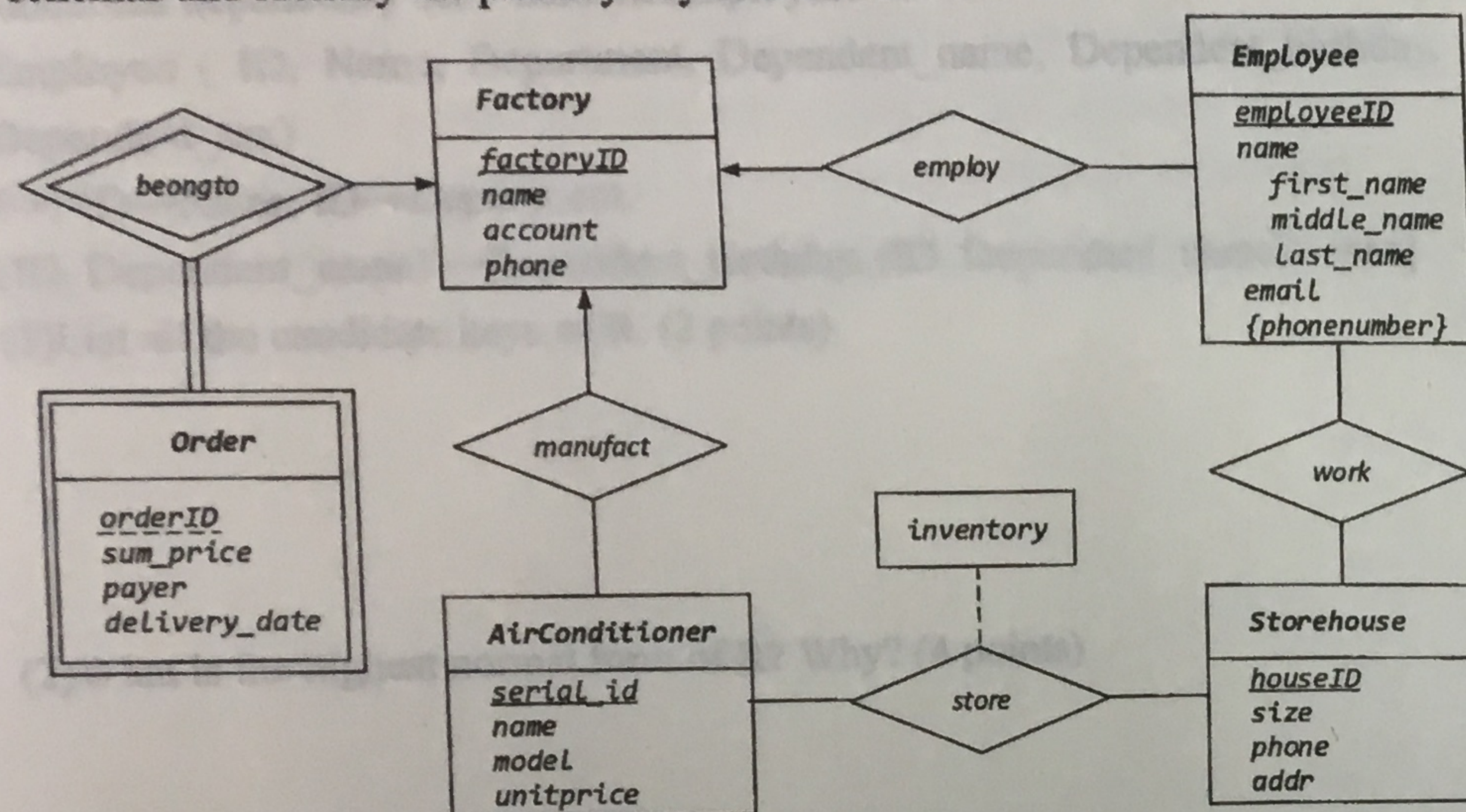


Fig. 2 E-R diagram

4. (10 points) Consider the following information of the database of a school sport-meeting management system:

- (1) Every Competition category is identified by a category_id and has a name and a manager.
- (2) Competition events are identified by event_id. For each event, the name,

competition time and level must be recorded.

- (3) Department teams are identified by team_number. Every team has a name, a leader.

- (4) Players are identified by player_id. For each player, the name, age, sex and phone_number must be recorded.

- (5) Referees are identified by referee_id. For each referee, the name, sex and department must be recorded.

- (6) Every competition category has several competition events. Each event belongs to a unique category.

- (7) Every team has several players. Each player belongs to a unique team.

- (8) Each player could attend different competition events. And each event can be attended by more than one player. Players have their grades in different events.

- (9) Each competition event has a referee, who judges the result.

Design the E-R Diagram on the basis of the information above.

5. (6 points) Consider the following relation schema Employee and the functional dependency set F hold on Employee:

Employee (ID, Name, Department, Dependent_name, Dependent_birthday, Dependent_sex)

F={ ID→Name, ID→Department, (ID, Dependent_name)→Dependent_birthday, (ID, Dependent_name)→sex}

(1) List all the candidate keys of R. (2 points)

(2) What is the highest normal form of R? Why? (4 points)

6. (8 points) The functional dependency set F={ A→D, E→D, D→B, BC→D, CD→A } holds on the relation schema R = (A, B, C, D, E)

(1) Is the decomposition $\rho = \{R_1(A, C, D), R_2(B, C, D, E)\}$ on R lossless-join and dependency preserving? Why? (4 points)

(2) Give a lossless-join and dependency-preserving decomposition of R into 3NF. (4 points)

7. (6 points) Indexing aims to speed up accessing data items in database. For the following three SQL queries and the indices defined on the tables.

(I) select *Student.ID*, *instructor.name*, *instructor.dept_name*,
from *instructor*, *student*
where *salary*>2000 and *instructor.dept_name*=*student.dept_name*

A nonclustered index defined on *salary* in the table *instructor*(*ID*, *name*, *dept_name*, *salary*), and a nonclustered index defined on *dept_name* in *student*(*name*, *ID*, *dept_name*, *age*, *sex*, *birthdate*);

(II) insert into *student*
values('Wang', 2113456, 'CS', 22, male, 19950505)

A clustered index on *ID* in the table *student*(*name*, *ID*, *dept_name*, *age*, *sex*, *birthdate*);

(III) update *student*
set *age*=*age*+1;
where *birthdate* between 19950619 and 19951022

A clustered index on *ID* in *student*(*name*, *ID*, *dept_name*, *age*, *sex*, *birthdate*);

(1) Which query can be speeded up by the index on the table it accesses? Why? (2 points)

(2) Which one will not benefit from the index at all, and why? (2 points)

(3) Which one is even slowed down by the index, and why? (2 points)

8. (11 points) Consider the **Banking** database given in Question Two.

(1) Give a SQL statement to find all customers, and list their names and the cities they reside in. It is required that the customer has **account** at the **branches** that are located in **Beijing** and have **assets** more than \$200,000, and the account is deposited before 20160101 and its **balance** is more than \$2000” (3 points)

(2) For the SQL statement in (1), give an optimized query tree by heuristic optimization. (8 points)

9. (6 points) Is S3 a cascadeless schedule? why?

S3:

T ₁	T ₂	T ₃	T ₄
	Read(Y)		
		read(M) M: =M+20	
			read(R) R: =R-20 read(Z) Z: =Z-30
	Y:= Y -20 write(Y) commit		
read(X) X: =X-20 write(X)			
			write(R) write(Z) commit
		write(M) read(Y) Y:= Y +5 write(Y) commit	
read(Z) Z: =Z+10 write(Z) commit			

10. (6 points) Concurrent schedule S is shown below.

(1) Construct the precedence graph for it. (3 points)

S:

T ₁	T ₂	T ₃	T ₄	T ₅
				read(Z) Z: =Z-10
		read(Y) Y: =Y*10 Write(Y)		
read(X) X: =X+50				
				write(Z)
Write(X)				
	read(X) X: =X-500			
				read(D)
	Write(X)			
		read(X) X: =X+50		
			read(Z)	
		Write(X)		
			Z: =Z*10	
				D: =D+10 write(D)
			write(Z)	
read(D) D: =D/5				
			read(Y) Y: =Y+20	
write(D)				
			Write(Y)	
	read(Q) Q: =Q+20 Write(Q)			

(2) Is S a serializable schedule? If not, give the reason. If it is, give a serial schedule equivalent to S. (3 points)

11. (11 points) (1) and (2) given below are about the database recovery.

(1) Fig.3 is the log of the concurrent executing of T₁, T₂, T₃ and T₄. It is assumed that the initial values of these data items in database are A=0, B=10, C=100, D=350.

(a) After every checkpoint, what are the values of the data items A, B, C, D in the database? (4 points)

```

<T1 start>
<T1, A, 0, 10>
<T2 start>
<T2, B, 10, 100>
<checkpoint {T1, T2}>
<T3 start>
<T3, C, 100, 200>
<T4 start>
<T4, D, 350, 500>
<T3, C, 200, 300>
<T3 commit>
<T2, B, 100, 80>
<checkpoint {T1, T2, T4}>
<T4, C, 300, 450>
<T4 commit>
<T1, A, 10, 100>
**failure**

```


Fig.3 Log of the concurrent executing of T₁, T₂, T₃ and T₄

(b) After recovery operations on T₁, T₂, T₃ and T₄ are completed, what are the values of the data items A, B, C, D in the database? (4 points)

(2) Fig. 4 is about a concurrent schedule of transactions T₁, T₂, T₃, T₄, T₅ and T₆. T_{c1}, T_{c2} and T_{c3} are the time of three times of checkpoint, T_f is the time of system failure. After the failure occurred, what recovery actions (i.e. redo, undo, ignore) should be conducted respectively for T₁, T₂, T₃, T₄, T₅ and T₆? (3 points)

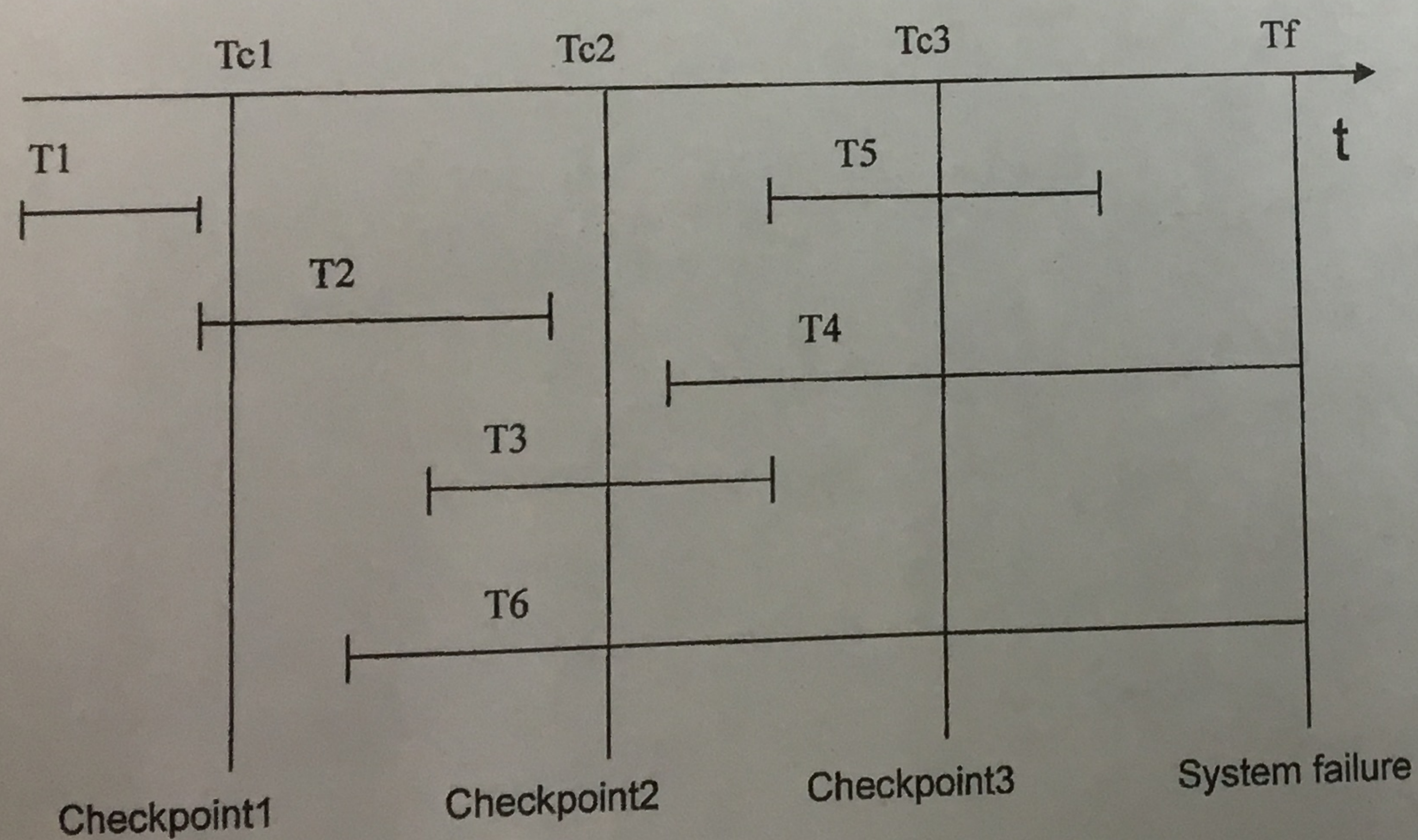


Fig.4 Concurrent schedule of transactions